

A Cleaned Global Earthquake Dataset for AddiVortes Benchmarking

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1 Introduction

This article documents a cleaned dataset of global earthquake events prepared for benchmarking the AddiVortes model. The dataset was chosen because it provides a large number of observations with a genuine mix of numerical covariates measured on fundamentally different scales, together with categorical covariates, and latitude/longitude coordinates spanning the entire globe. The latter is particularly useful for testing distance-based splitting rules, since great-circle distance is the natural metric for the coordinate pair while Euclidean distance is appropriate for the remaining numerical covariates.

2 Data Source

The raw data was obtained from the U.S. Geological Survey’s Earthquake Hazards Program, via the public ANSS Comprehensive Earthquake Catalog (ComCat) query API:

<https://earthquake.usgs.gov/earthquakes/search/>

The pull used for this article consists of the most recent 20,000 earthquake records in the catalog which occurred in 2025 (magnitude ≥ 2.5), which corresponds to events occurring between 5 May 2025 and 31 December 2025. The USGS API caps any single query at 20,000 returned rows; a full multi-decade pull requires paginating over the `offset` parameter or splitting the request into shorter date ranges, but the analysis in this article is restricted to this single 20,000-row pull as a representative working sample.

The raw file, `raw_data.csv`, contained 20,000 rows and 19 columns: `time`, `latitude`, `longitude`, `depth`, `mag`, `magType`, `nst`, `gap`, `dmin`, `rms`, `net`, `type`, `horizontalError`, `depthError`, `magError`, `magNst`, `status`, `locationSource`, and `magSource`.

3 Data Cleaning

The raw data required six cleaning steps: (i) restricting to genuine earthquake events, since the catalog also contains a small number of mining explosions and other non-tectonic events; (ii) removing columns that were pure metadata, near-duplicates of another retained column, or deliberately excluded from the benchmark covariate set; (iii) removing the small proportion of rows with residual missing values in the quality-control columns; (iv) subsetting to reviewed body-wave magnitude events, i.e. `magType == "mb"` and `status == "reviewed"`; and (v) dropping the now-constant

`magType` and `status` columns from the retained covariate set. The full cleaning pipeline, implemented in R, is reproduced below.

Listing 1: R script used to clean the raw USGS earthquake catalog extract.

```

1 # =====
2 # AddiVortes test data prep: USGS earthquake catalog cleanup
3 # =====
4
5 # ---- 1. Read in the raw CSV ----
6 df <- read.csv("raw_data.csv", stringsAsFactors = FALSE)
7
8 cat("Starting rows:", nrow(df), "\n")
9 cat("Starting columns:", ncol(df), "\n\n")
10
11 # ---- 2. Filter to real earthquakes only ----
12 # 'type' includes a small number of mining explosions, landslides, etc.
13 # which are noise for an earthquake-magnitude model.
14 before_n <- nrow(df)
15 df <- df[df$type == "earthquake", ]
16 cat("After filtering to type == 'earthquake':", nrow(df),
17     "rows (dropped", before_n - nrow(df), ")\n")
18
19 # ---- 3. Drop redundant / non-covariate columns ----
20 # - 'type' is now constant (all 'earthquake') -> no information left
21 # - 'locationSource' is ~99.99% identical to 'net' -> redundant
22 # - 'magSource' is ~99.7% identical to 'net' -> redundant
23 # - 'time' dropped for this non-time-series test (re-add + engineer
24 #   year/month/hour if you want temporal covariates later)
25 # - 'gap' removed from the benchmark covariate set as requested
26 drop_cols <- c("type", "locationSource", "magSource", "time", "gap")
27 df <- df[, !(names(df) %in% drop_cols)]
28 cat("Dropped columns:", paste(drop_cols, collapse = ", "), "\n")
29 cat("Remaining columns:", paste(names(df), collapse = ", "), "\n\n")
30
31 # ---- 4. Drop rows with any remaining missing values ----
32 # Missingness across the numeric quality/error columns is low (~5.6%)
33 # and concentrated in a subset of 'ml' and 'mw' events, so row deletion
34 # is cheap and keeps the dataset fully complete.
35 before_n <- nrow(df)
36 df_clean <- df[complete.cases(df), ]
37 dropped_n <- before_n - nrow(df_clean)
38 cat("After dropping rows with missing values:", nrow(df_clean),
39     "rows (dropped", dropped_n,
40     sprintf("%.1f%%]", 100 * dropped_n / before_n), ")\n\n")
41
42 # ---- 5. Restrict to the benchmark subset used in this session ----
43 # Keep only reviewed body-wave magnitude measurements for the benchmark.
44 before_n <- nrow(df_clean)
45 df_clean <- df_clean[df_clean$magType == "mb" & df_clean$status == "
46     reviewed", ]
47 cat("After filtering to magType == 'mb' and status == 'reviewed':",
48     nrow(df_clean), "rows (dropped", before_n - nrow(df_clean), ")\n\n")
49
50 # ---- 6. Drop now-constant filtering columns ----

```

```

50 # After subsetting, 'magType' and 'status' are constant and carry no
    information.
51 df_clean <- df_clean[, !(names(df_clean) %in% c("magType", "status"))]
52 cat("After dropping constant columns magType and status:",
53     nrow(df_clean), "rows and", ncol(df_clean), "columns\n\n")
54
55 # ---- 7. Sanity checks ----
56 cat("Final shape:", nrow(df_clean), "rows x", ncol(df_clean), "columns\n")
57 cat("Total missing values remaining:", sum(is.na(df_clean)), "\n\n")
58
59 cat("Categorical cardinality:\n")
60 for (col in c("net")) {
61   cat(" ", col, "->", length(unique(df_clean[[col]])), "levels\n")
62 }
63
64 # ---- 8. Write out the cleaned file ----
65 write.csv(df_clean, "earthquakes.csv", row.names = FALSE)
66 cat("\nSaved cleaned file to earthquakes.csv\n")

```

Applying this pipeline reduced the raw extract from 20,000 rows and 19 columns to a final cleaned dataset of 11,697 rows and 12 columns, with zero remaining missing values. Table 1 summarises each stage.

Stage	Rows	Columns
Raw extract	20,000	19
After filtering to <code>type == "earthquake"</code>	19,841	19
After dropping redundant columns	19,841	14
After dropping incomplete rows	18,732	14
After subsetting to <code>magType == "mb"</code> and <code>status == "reviewed"</code>	11,697	14
After dropping constant <code>magType</code> and <code>status</code> columns	11,697	12

Table 1: Row and column counts at each stage of the cleaning pipeline.

4 Covariate Descriptions

Table 1 describes each of the 12 covariates retained in the final cleaned dataset, along with its type and the range (for numeric variables) or levels (for categorical variables) observed in the data.

Covariate	Type	Description	Range / Levels
latitude	Numeric	Epicenter latitude in decimal degrees (WGS84). Natural great-circle covariate.	-65.6° to 87.1°
longitude	Numeric	Epicenter longitude in decimal degrees (WGS84). Natural great-circle covariate.	-180.0° to 180.0°

Covariate	Type	Description	Range / Levels
<code>depth</code>	Numeric	Depth of the hypocenter below the surface, in kilometers. Small negative values reflect very shallow events located slightly above the reference datum.	−3.4 to 669.6 km
<code>mag</code>	Numeric	Earthquake magnitude, on the body-wave scale used for the retained events. Natural target variable.	2.5 to 8.8
<code>nst</code>	Numeric	Number of seismic stations used to determine the event’s <i>location</i> .	3 to 566
<code>dmin</code>	Numeric	Horizontal distance, in degrees, from the epicenter to the nearest reporting station.	0 to 55.9
<code>rms</code>	Numeric	Root-mean-square of the travel-time residuals, in seconds; a location goodness-of-fit measure.	0 to 2.0 s
<code>net</code>	Categorical	Seismic network that contributed the preferred origin (e.g. national or regional networks).	12 levels, dominated by <code>us</code> (16,088 rows)
<code>horizontalError</code>	Numeric	95% confidence radius on the epicenter location, in kilometers.	0 to 40.8 km
<code>depthError</code>	Numeric	95% confidence uncertainty on the depth estimate, in kilometers.	0 to 42.9 km
<code>magError</code>	Numeric	Standard error on the magnitude estimate, in magnitude units.	0 to 1.3
<code>magNst</code>	Numeric	Number of seismic stations used to compute the <i>magnitude</i> (distinct from <code>nst</code>).	0 to 1,027

Table 2: Covariates in the cleaned earthquake dataset.

5 Notes for Model Testing

One covariate warrants caution during testing: `net` is highly imbalanced, with the `us` network accounting for roughly 80% of observations; this may limit the effective diversity of that categorical covariate for split-finding. Aside from that variable, the dataset offers a genuine mix of scales and types — degrees, kilometers, seconds, magnitude units, and station counts among the numeric covariates, together with a single categorical covariate — with global latitude/longitude coverage and no missing data, making it well suited to benchmarking AddiVortes against both great-circle and Euclidean distance metrics.